

Education

Stony Brook University	2019–present
Ph.D. student, Electrical and Computer Engineering. Research focus: neuromorphic signal processing and EEG/ERP representation analysis. Advisor: Wendy Tang; committee/collaborator: Brady D. Nelson.	
New York Institute of Technology	2010
M.S., Electrical and Computer Engineering.	
Hofstra University	2008
B.S., Biomedical and Electrical Engineering.	

Research Experience

Doctoral Researcher, Stony Brook University	2019–Present
Developed ARSPI-Net, a staged neuromorphic signal-processing framework for EEG/ERP analysis. Doctoral work models affective neural data as noisy, partially observed outputs of biological dynamical systems.	
Bioinformatics Analyst, Feinstein Institute for Medical Research	2011–2014
Developed laboratory-automation robotics software and research data-management systems.	
Research Analyst, Rockefeller University	2008–2011
Developed automation and robotics programming for high-throughput screening workflows.	

Teaching and Academic Appointments

Teaching Professor, St. Joseph’s University	2019–Present
Department of Mathematics and Computer Science. Teaching areas include artificial intelligence, data structures and algorithms, operating systems, computer networking, information security, computer forensics, programming, discrete mathematics, and computer architecture.	
Adjunct Instructor, St. Joseph’s University	2014–2019
Department of Mathematics and Computer Science.	
Adjunct Instructor, Hofstra University	2017–Present
Computer science and engineering instruction.	
Adjunct Assistant Professor, Farmingdale State College	2014–Present
Computer science and engineering instruction.	
Instructor, Stony Brook University	2022–2025
Department of Electrical Engineering.	

Selected Publications

- A. Lane, W. Tang, and B. Nelson, “Towards ARSPI-Net: Advancing EEG Feature Extraction with Neuromorphic Algorithms,” *2024 IEEE Long Island Systems, Applications and Technology Conference (LISAT)*, pp. 1–8, 2024. doi:10.1109/LISAT63094.2024.10808138.
- A. Lane, W. Tang, and B. Nelson, “Towards ARSPI-Net: Development of an Efficient Hybrid Deep Learning Framework,” *2023 IEEE Long Island Systems, Applications and Technology Conference (LISAT)*, pp. 1–8, 2023. doi:10.1109/LISAT58403.2023.10179592.

Mentoring and Service

- Faculty advisor, Upsilon Pi Epsilon Computer Science Honor Society, St. Joseph’s University.
- Undergraduate research mentor for projects in neural machine translation, visual speech recognition, affective computing, and AI bias.

Technical Skills

Neuromorphic computing; spiking neural networks; liquid-state machines; reservoir computing; graph neural networks; EEG/ERP signal processing; perturbation analysis; dimensionality reduction; Python; machine learning; LaTeX.